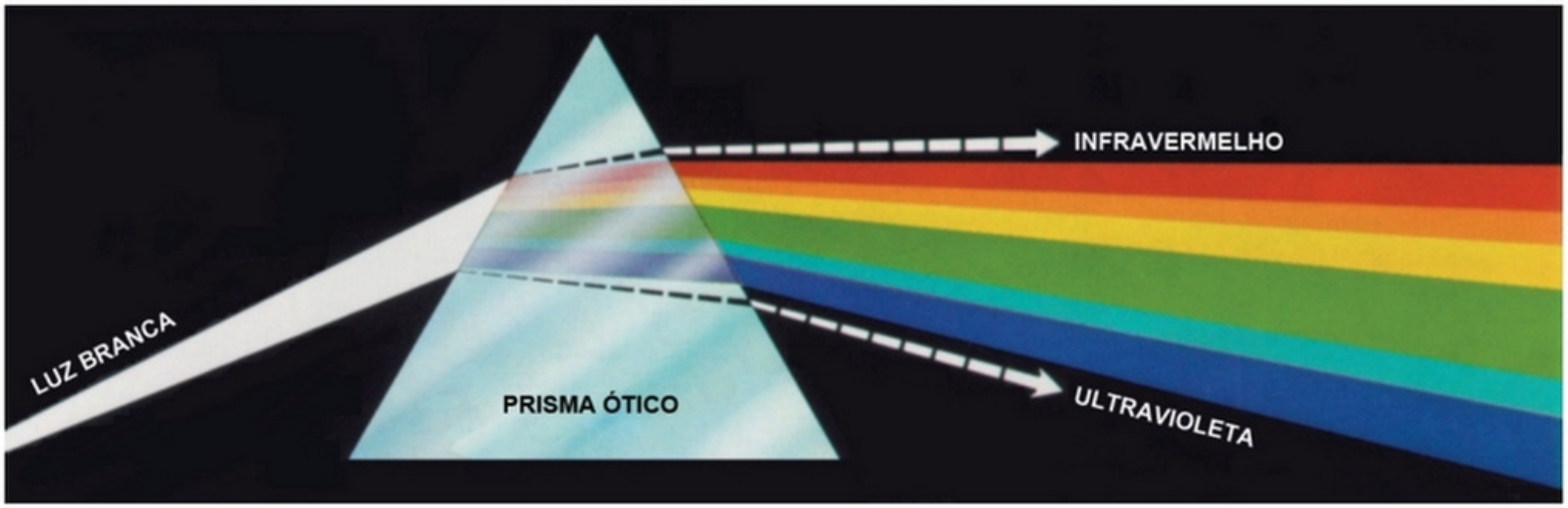
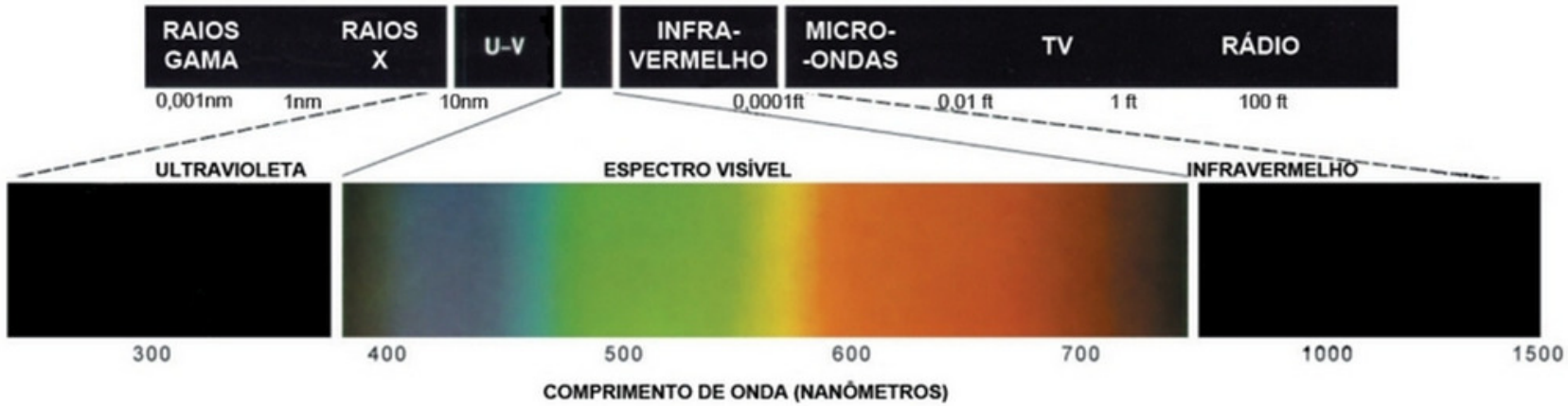
6. PROCESSAMENTO DE IMAGENS COLORIDAS

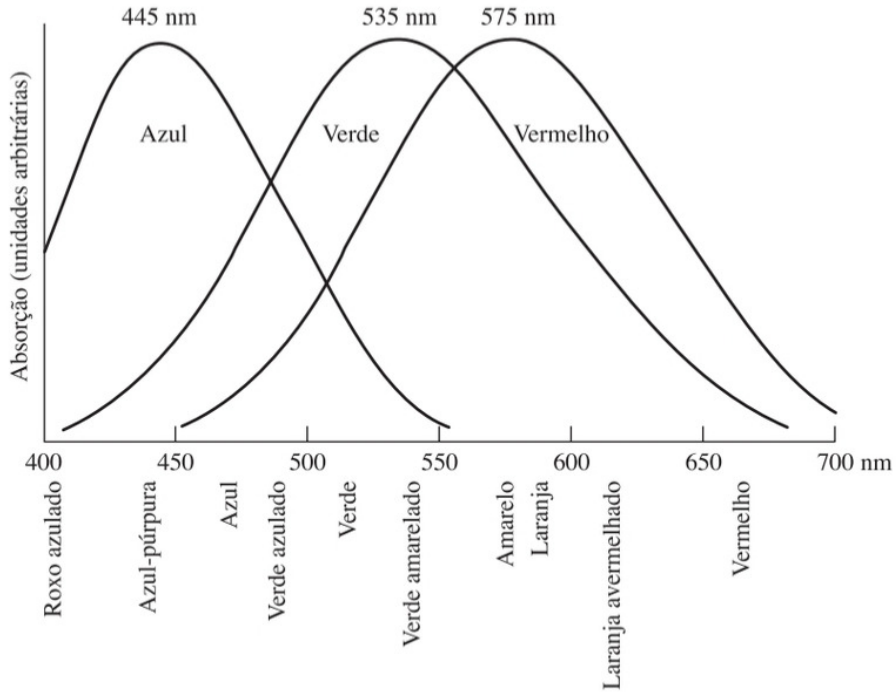
6.1. FUNDAMENTOS DAS CORES



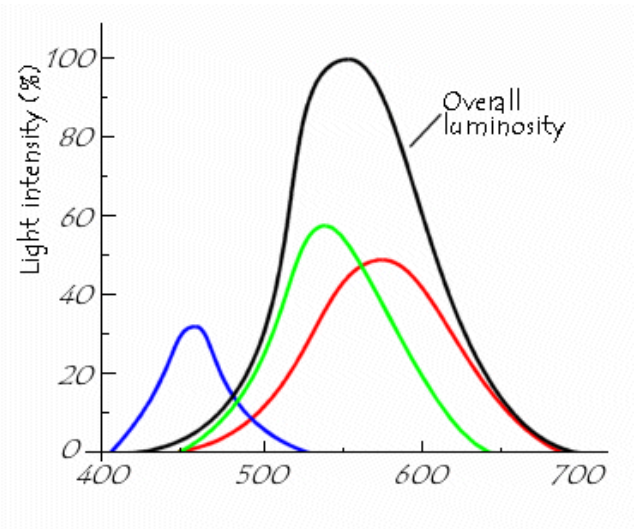


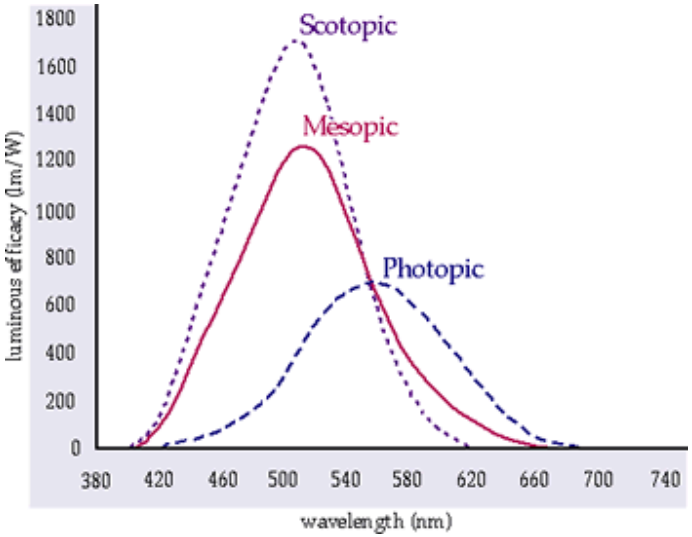
- Cor de um objeto
  - Objetos "brancos" refletem todos os comprimentos de onda
  - Objetos "vermelhos" refletem os comprimentos de onda na faixa do vermelho
  - ...

- Percepção de uma cor
  - Cones
    - R (65%)
    - G (33%)
    - B (2%)



- Sensibilidade frente a condições média de iluminação



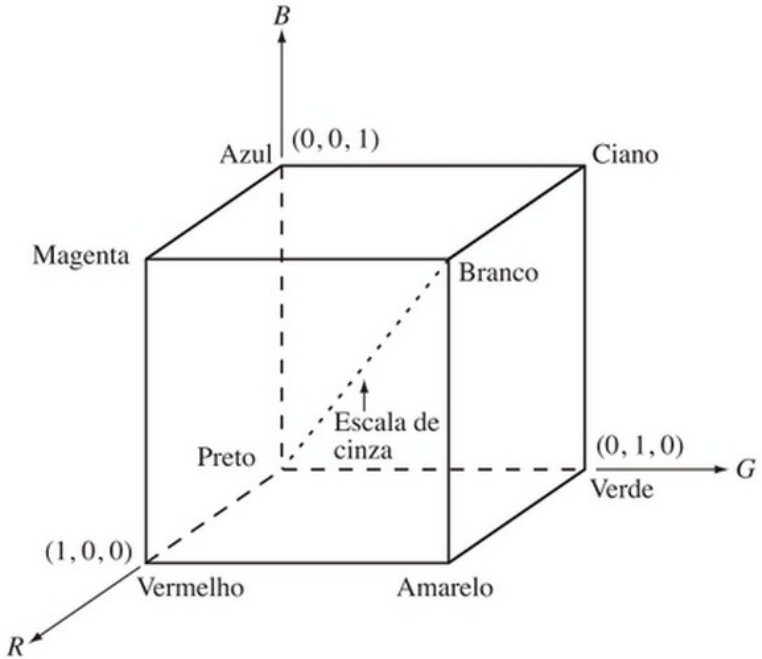


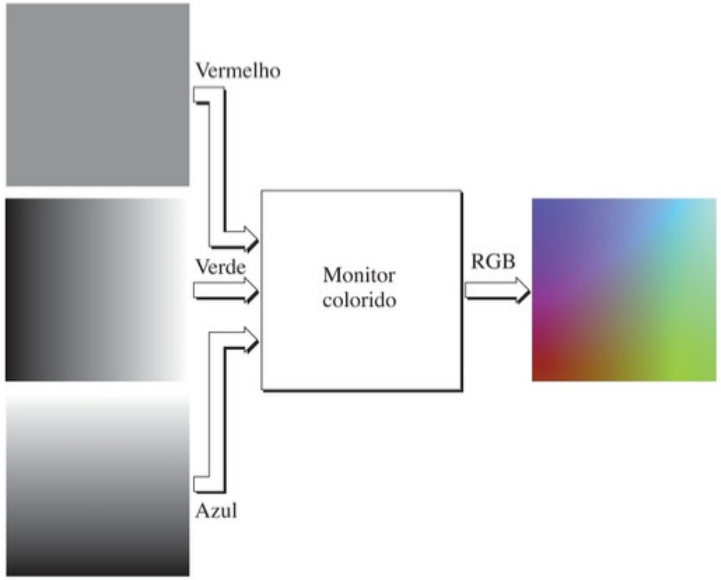
- Características (comuns) "cor"
  - Brilho: noção acromática de intensidade
  - Matiz: atributo associado ao comprimento de onda dominante
  - Saturação: quantidade de luz branca na mistura

- Matiz + Saturação $\Rightarrow$ Cromaticidade
- Cor $\Rightarrow$ Brilho + Cromaticidade

6.2. MODELOS DE CORES (ESPAÇOS/SISTEMAS DE CORES)

- RGB

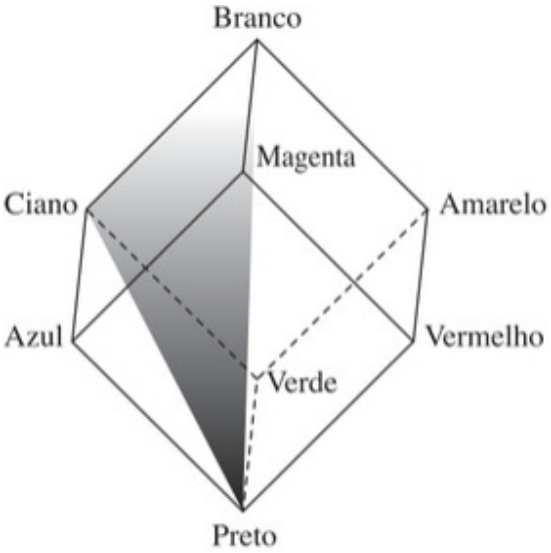


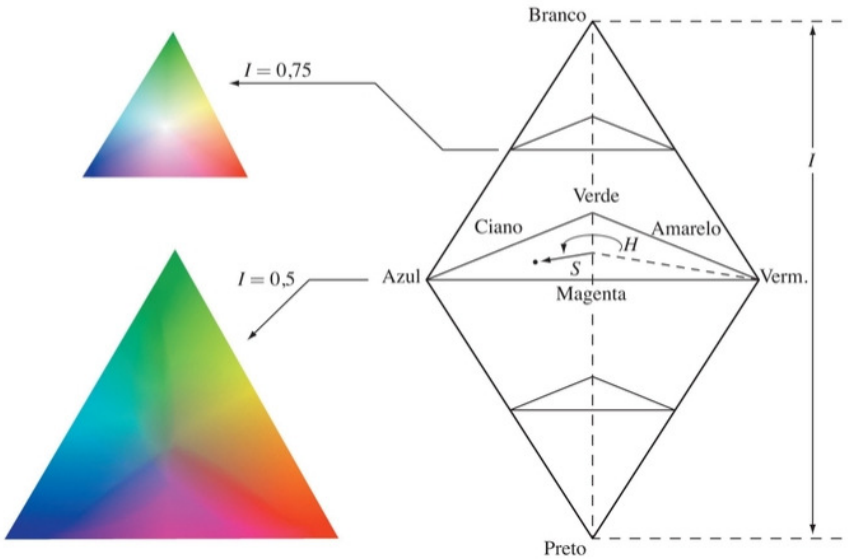


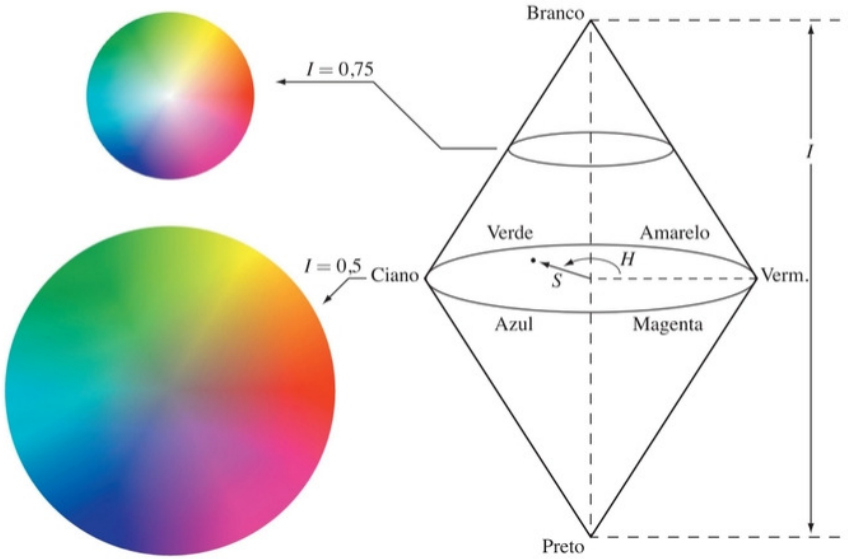
- CMY
  - Utilizado por sistemas de impressão

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- HSI


Branco
Magenta
Ciano
Amarelo
Azul
Vermelho
Verde
Preto


Branco
I = 0,75
I
Verde
Ciano
Amarelo
H
S
I = 0,5
Azul
Verm.
Magenta
Preto


Branco
I = 0,75
I
Verde
Amarelo
H
I = 0,5
Ciano
S
Verm.
Azul
Magenta
Preto

$$H = \begin{cases} \theta & \text{se } B \le G \\ 360 - \theta & \text{se } B > G \end{cases} \quad ; \quad \theta = \cos^{-1}\left\{ \frac{\frac{1}{2}[(R-G)+(R-B)]}{[(R-G)^2+(R-B)(G-B)]^{1/2}} \right\}$$

$$S = 1 - \frac{3}{R+G+B}[\min(R,G,B)]$$

$$I = \frac{1}{3}(R+G+B)$$

- Conversão HSI para RGB
  - $0^\circ \le H < 120^\circ$ (Setor RG)

$$B = I(1-S) \quad ; \quad R = I\left[1 + \frac{S\cos(H)}{\cos(60^\circ - H)}\right] \quad ; \quad G = 3I - (R+B)$$

- $120^\circ \leq H < 240^\circ$ (Setor GB)

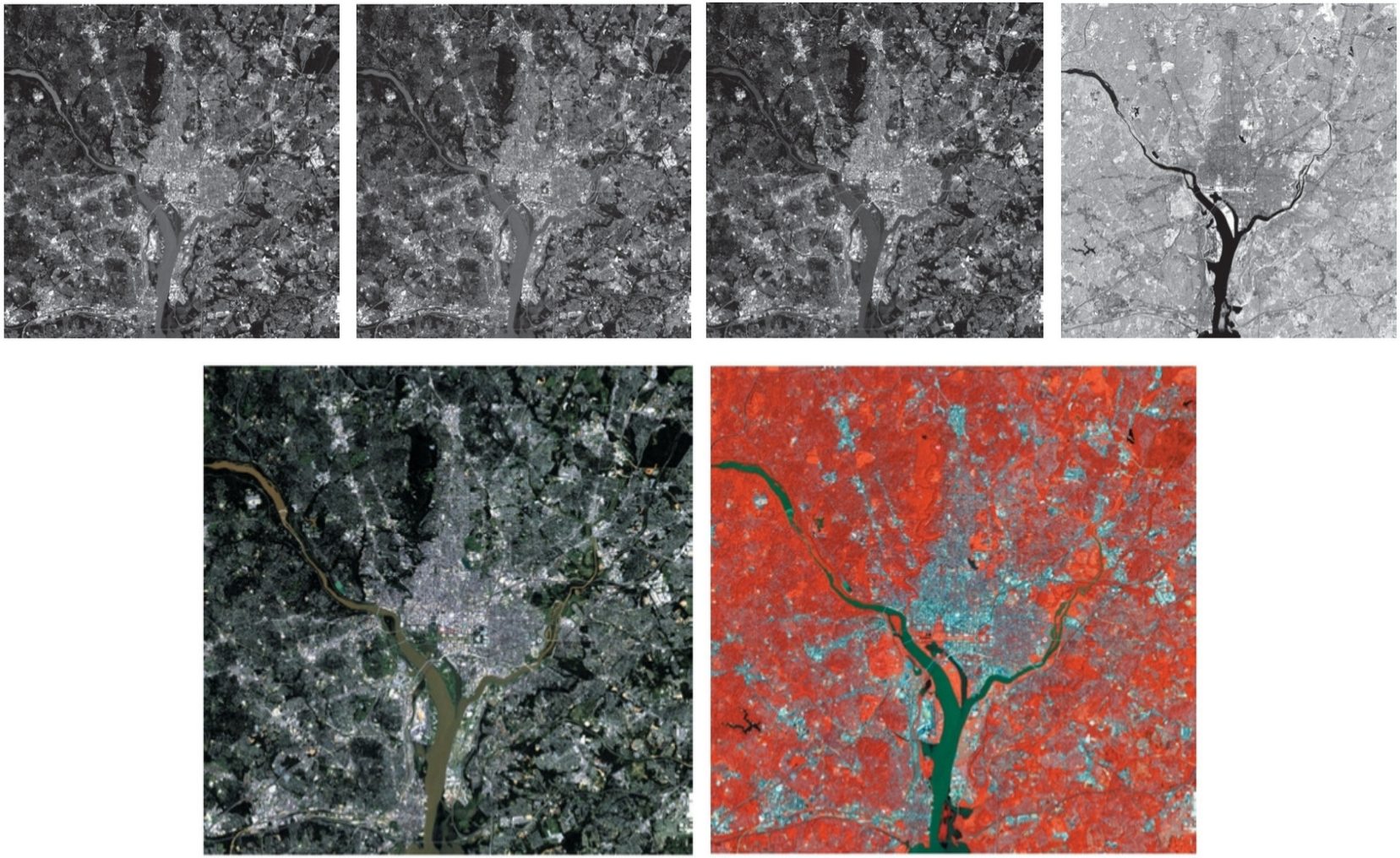
$$H = H - 120^\circ$$

$$R = I(1-S) \quad ; \quad G = I\left[1 + \frac{S\cos(H)}{\cos(60^\circ - H)}\right] \quad ; \quad B = 3I - (R+G)$$

- $240^\circ \leq H < 360^\circ$ (Setor BR)

$$H = H - 240^\circ$$

$$G = I(1-S) \quad ; \quad B = I\left[1 + \frac{S\cos(H)}{\cos(60^\circ - H)}\right] \quad ; \quad R = 3I - (G+B)$$

- Outros Modelos
  - Intensidade: $I = \frac{1}{3}(R+G+B)$
  - Valor: $V = \max(R,G,B)$
  - Iluminação: $L = 0.5 \cdot \max(R,G,B) + 0.5 \cdot \min(R,G,B)$
  - Luminânica: $Y = 0{,}30R + 0{,}59G + 0{,}11B$

6.3. IMAGENS MULTIESPECTRAIS